

Minor in Robotics

Minor Details

Minor name	Minor in Robotics
Eligibility batch(es)	Batches 2023, 2024

Table 1: Minor in Robotics — Requirements

Sl.	Course code	Course name	L-T-P-C	Credits
Compulsory: Robot Kinematics / Dynamics / Control All 1 listed course required; Counts toward minor credits				
1	AR-501 / ME-452	Robot Kinematics, Dynamics, and Control	3-1-0-4	4
Compulsory: Mechatronics (pick 1) Choose 1 of 2; Counts toward minor credits				
2	AR-503	Mechatronics	3-0-0-3	3
3	AR-520	Design Practicum of Mechatronic Systems	—	3
Electives Choose 2 of 15; Counts toward minor credits				
4	AR-502	Advanced Design Practicum	3-0-2-4	4
5	AR-504	Robot Programming	2-0-2-3	3
6	AR-505	Principles of Robot Autonomy	3-0-0-3	3
7	AR-506	Cognitive Robotics	3-0-0-3	3
8	AR-507	Probabilistic Robotics	3-0-0-3	3
9	AR-508	Marine Robotics	3-0-0-3	3
10	AR-509	Deep Learning for Robotics	3-0-2-4	4
11	AR-510	Underactuated Robotics	3-0-0-3	3
12	AR-511	Autonomous Mobile Robots	3-0-0-3	3
13	AR-512	Rapid Prototyping and Tooling	3-0-2-4	4
14	AR-513	Unmanned Aerial Systems (UAS)	3-0-0-3	3
15	AR-514	Vision and Learning Based Control	3-0-0-3	3
16	AR-515	Sensors and State Estimation	3-0-0-3	3
17	AR-519	Robot Manipulators	—	3
18	EE-555	Intelligent Control System	3-0-0-3	3

Requirement notes

- **Compulsory: Mechatronics (pick 1):** AR-503 or AR-520 (equivalent courses — complete either one).
- **Electives:** Any 2 out of the electives list (as per curriculum).